SEs for Cluster Randomized Experiments

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Overview

In this report, I examine when the superpopulation and finite population standard errors can differ massively, and find that they bascially cannot for most reasonable parameters of interest. I don't look at variable cluster sizes, but I think that will have little to no impact on the findings.

I do some math derivations under a simple model, then a model where there is correlation of site means and site impacts. I check the math with a small set of simulations, and the simulations seem to align.

The short version of the overall finding is that to have a large difference term between the superpopulation and finite population standard errors, you need to have a lot of treatment effect variation, but when you do, the finite sample standard errors also tend to grow, making the ratio of super to finite not as large as one might expect.

Math Derivations

Let μ_j be the average control-side mean and ATE_j be the average treatment effect of cluster j.

Take a DGP with a given ICC and τ^2, where τ is the standard deviation of the cluster average effects: $\tau^2 = Var(ATE_j)$. Further assume all our clusters are the same size, $n_j = n$.

Now take a simple difference in means estimator for an estimate of the cluster-average impact:

$$\widehat{ATE} = \frac{1}{pJ}\sum_{j=1}^{J} Z_j \bar{Y}_j - \frac{1}{(1-p)J}\sum_{j}(1-Z_j)\bar{Y}_j$$

where p is the proportion of clusters treated.

The finite-sample Neyman variance of this estimator is:

$$\text{Var}(\widehat{ATE}) = \frac{1}{J_1}Var(\bar{Y}_j(1)) + \frac{1}{J_0}Var(\bar{Y}_j(0)) - \frac{1}{J}Var(ATE_j),$$

where $Var(\bar{Y}_j(z))$ is the variance of the cluster average effects under given treatment arm z.

Under a superpopulation framing, we can think of each arm as independently drawn from a larger population, with

$$Var(\bar{Y}_j(0)) = \left(ICC + \frac{(1-ICC)}{n_j}\right)\sigma^2$$

and

$$Var(\bar{Y}_j(1)) = \tau^2 + \left(ICC + \frac{(1-ICC)}{n_j}\right)\sigma^2$$

This is under the assumption that the cluster treatment effects are independent of the cluster means (so we can add the ICC and τ^2 terms). This may be a very important thing to ponder.

Under effect size units, drop the σ^2 and we have a superpopulation variance (SE squared) of

$$\begin{aligned}
\text{Var}(\widehat{ATE})_{super} &= \frac{1}{J_1} Var(\bar{Y}_j(1)) + \frac{1}{J_0} Var(\bar{Y}_j(0)) \\
&= \frac{1}{J_1}\left(ICC + \frac{(1-ICC)}{n}\right) + \frac{1}{J_1}\tau^2 + \frac{1}{J_0}\left(ICC + \frac{(1-ICC)}{n}\right) \\
&= \frac{ICC}{p(1-p)J} + \frac{(1-ICC)}{p(1-p)Jn} + \frac{1}{J_1}\tau^2
\end{aligned}$$

For finite SEs, we have this but have to subtract off of the variance of the ATE to get

$$SE^2_{finite} = \text{Var}(\widehat{ATE})_{finite} = \frac{ICC}{Jp(1-p)} + \frac{(1-ICC)}{p(1-p)Jn} + \left(\frac{1}{J_1} - \frac{1}{J}\right)\tau^2$$

So superpopulation SEs will not generally be substantially different, if the cluster treatment effects are uncorrelated with control-side cluster means. In particular, $Var_{super} - Var_{finite} = \frac{1}{J}\tau^2$ and the percent increase of super vs. finite would be

$$\frac{Var_{super} - Var_{finite}}{Var_{finite}} = \frac{p \cdot \tau^2}{\left(\frac{ICC}{1-p} + \frac{1}{(1-p)n} + \tau^2\right)}.$$

To make superpopulation SEs large, relative to finite SEs, we need τ^2 to be large, n to be large, and ICC to be small. Interestingly, if p is large, then $1-p$ is small, which makes the ratio even larger.

Overall, however, what is happening is that in order to increase heterogeneity in treatment effects, we are also increasing heterogeneity in the treatment group, and elevating the standard error. The relative size, therefore, is not changing much.

As a numerical example, say $J = 10$, and $\tau^2 = 0.3^2$. Then $\tau^2/J = 0.3^2/10 = 0.01$, a small increase in the superpopulation variance vs. finite. (Take the square root to get 0.10, but this is not actually the increase as it would be $\sqrt{SE^2_{fin} + 0.01}$, which is likely to be close to SE_{fin}. E.g., if $SE_{fin} = 0.20$ we would get around 0.22, a modest increase.)

So how could they be different?

All of the above was under the assumption that the cluster mean under control was uncorrelated with the cluster average impact. If they are not correlated, then the variance of the $\bar{Y}_j(1)$ is no longer $ICC + \tau^2$, but instead $ICC + \tau^2 + cov(ICC, ATT_j)$.

We can therefore make the finite population SE smaller by making the correlation between the cluster average impact and the cluster average control mean negative. The smaller it gets, the larger the ratio of superpopulation to finite SEs.

In particular, we have

$$Var(\bar{Y}_j(1)) = \tau^2 + \left(ICC + \frac{(1-ICC)}{n_j}\right)\sigma^2 + \sqrt{\tau^2 \cdot ICC}\rho$$

where $\rho = cor(ATE_j, \mu_j)$. We only have correlation on the cluster mean, not the noise of individual students in the cluster, hence the final term not having the $1/n$ term in it.

This gives

$$\frac{Var_{super} - Var_{finite}}{Var_{finite}} = \frac{p \cdot \tau^2}{\left(\frac{ICC}{1-p} + \frac{1-ICC}{(1-p)n} + \tau^2 + \sqrt{\tau^2 \cdot ICC}\rho\right)}$$

(I think the above holds–this is careless math here, and I am operating on intuition, not actual derivation.)

To make this large, we need $ICC > 0$ so the ρ can do its thing. But if ICC is too large, then our ratio will be small anyway.

To discover the ideal balancing points, we can set up an optimizer! In the following code I implement the above formula and then search across a range of parameters to maximize the ratio.

```
## [1] -1.108386
```

In the following we optimize over ICC, tau2, p, n, and rho to find the largest ratio. I do not include J since it is not in the formula of the ratio. I flip the sign of the ratio as the optimizer looks for a minimum—we want the maximum. I do not know if our problem is convex, so it is possible we are seeing a local maximum rather than global, but we can fiddle with start parameters to try and find something different if needed.

We also bound the parameter values between lower and upper bounds.

Here is the initial stab:

```
pp = c( ICC = 0.2, tau2 = 0.2, p = 0.75, n = 1, rho = -1 )
opt <- optim(  pp, a,
               method ="L-BFGS-B",
               lower = c(0, 0, 0.01, 1, -1),
               upper = c(1, 200, 0.99, 10000,   1))

a( opt$par )
```

```
## [1] -8.124652
```

```
round( opt$par, digits = 2 )
```

```
##    ICC   tau2      p      n    rho
##   0.01 200.00   0.99  64.37  -1.00
```

We see that if most of our units are treated, then we get our most extreme difference.

If we constrain $p = 7/9$ and $n = 500$ we get

```
##  ICC tau2  rho
##    1   81   -1
```

and SEs of

super	finite	vY1	vY0	ratio
3.31	1.39	73	1	2.38

Here, to get a large ratio, we max out the ICC, have an incredibly large treatment variation, and have a strong negative correlation.

If we limit the amount of treatment variation to at most 1 we get:

```
##   ICC  tau2   rho
##  0.01  1.00 -1.00
```

and SEs and ratio of

super	finite	vY1	vY0	ratio
0.37	0.16	0.9	0.01	2.33

We end up a ratio of 233%, i.e., our superpopulation SE is 2.3 times as large. So we can get big, if our treatment variation is huge.

If we constrain $tau2 = 0.5^2$ (which is still huge) we get

```
##      ICC    tau2     rho
##   0.0031  0.2500 -1.0000
```

and SEs of

super	finite	vY1	vY0	ratio
0.19	0.09	0.23	0.01	2.2

If we constrain $\tau^2 = 0.2^2$ we get

```
##       ICC     tau2      rho
##   0.00074  0.04000 -1.00000
```

and SEs of

super	finite	vY1	vY0	ratio
0.08	0.05	0.04	0	1.72

Comparing to multi-site experiments

For multisite experiments, we empirically found superpopulation SEs to be around 20% larger than finite, *on average*. But we also found contexts where there was great difference. In multisite experiments we have

$$SE^2 = \frac{\tau^2}{J} + \frac{1 - ICC}{Jp(1-p)n}$$

and the ratio of super to finite was

$$\frac{\frac{\tau^2}{J}}{\frac{1-ICC}{Jp(1-p)n}} = \frac{\tau^2 p(1-p)n}{1-ICC}$$

If $p = 0.5, n = 100$, and $ICC = 0.2$, this gives $\approx 31\tau^2$. If $\tau^2 = 0.3^3$, for example, this is 2.8, which is a large difference. We do not seem to be able to achieve such differences with cluster RCTs.

A simulation to check

Let's simulate a simple case where we have a DGP with a given ICC and τ^2 and see if our formula are correct. We start with $J = 9, n = 500, ICC = 0, \tau^2 = 0.3^2, p = 7/9$ and $\rho = 0$. The math says:

super	finite	vY1	vY0	ratio
0.12	0.06	0.09	0	1.85

And the simulation says:

method	weight	mean.SE.hat	bias	MCSE.cover	SE.s	SE.f	MCSE.SE	ratio
LRa-db	Cluster	0.11	0	0	0.12	0.06	0.02	1.91
LRa-het	Cluster	0.11	0	0	0.12	0.06	0.02	1.91
LRapw-db	Person	0.11	0	0	0.12	0.06	0.02	1.91

method	weight	mean.SE.hat	bias	MCSE.cover	SE.s	SE.f	MCSE.SE	ratio
LRapw-het	Person	0.11	0	0	0.12	0.06	0.02	1.91
LRi-crve	Person	0.11	0	0	0.12	0.06	0.02	1.91
LRi-db	Person	0.11	0	0	0.12	0.06	0.02	1.91
LRicw-db	Cluster	0.11	0	0	0.12	0.06	0.02	1.91
MLM	Cluster	0.21	0	0	0.12	0.06	0.02	1.91

They seem to match! (At least in this case.)

When I set the correlation of impact to level to -1, following our second to last optimization example from above (with $\tau^2 = 0.5^2$), I get this:

method	weight	mean.SE.hat	bias	MCSE.cover	SE.s	SE.f	MCSE.SE	ratio
LRa-FE-db	Cluster	0.22	-0.01	0.01	0.22	0.21	0.04	1.07
LRa-FE-het	Cluster	0.22	-0.01	0.01	0.22	0.21	0.04	1.07
LRa-FIcw-db	Cluster	0.22	-0.01	0.01	0.22	0.21	0.04	1.07
LRa-FIcw-het	Cluster	0.22	-0.01	0.01	0.22	0.21	0.04	1.07
LRapw-FE-db	Person	0.23	-0.01	0.01	0.23	0.21	0.04	1.08
LRapw-FE-het	Person	0.23	-0.01	0.01	0.23	0.21	0.04	1.08
LRapw-FIpw-db	Person	0.22	-0.01	0.01	0.23	0.21	0.04	1.08
LRapw-FIpw-het	Person	0.22	-0.01	0.01	0.23	0.21	0.04	1.08
LRi-FE-crve	Person	0.23	-0.01	0.01	0.23	0.21	0.04	1.08
LRi-FE-db	Person	0.23	-0.01	0.01	0.23	0.21	0.04	1.08
LRi-FIpw-crve	Person	0.22	-0.01	0.01	0.23	0.21	0.04	1.08
LRi-FIpw-db	Person	0.22	-0.01	0.01	0.23	0.21	0.04	1.08
LRicw-FE-db	Cluster	0.22	-0.01	0.01	0.22	0.21	0.04	1.07
LRicw-FIcw-db	Cluster	0.22	-0.01	0.01	0.22	0.21	0.04	1.07
MLM-FE	Cluster	0.21	-0.01	0.01	0.22	0.21	0.04	1.07
MLM-FIcw	Cluster	0.21	-0.01	0.01	0.22	0.21	0.04	1.07
MLM-RE	Cluster	0.21	-0.05	0.01	0.22	0.21	0.04	1.07

The numbers are coming in low, and the ratio is higher than expected. But the SEs are large enough this could be noise.

One more, for luck, with $ICC = 0.05$, $\tau^2 = 0.3^2$, $J = 16$, $n = 100$, $p = 3/4$ and $\rho = -0.7$.

The math says:

super	finite	vY1	vY0	ratio
0.15	0.13	0.1	0.06	1.15

The simulation says:

method	weight	mean.SE.hat	bias	MCSE.cover	SE.s	SE.f	MCSE.SE	ratio
LRa-db	Cluster	0.13	0	0.01	0.14	0.12	0.03	1.2
LRa-het	Cluster	0.13	0	0.01	0.14	0.12	0.03	1.2
LRapw-db	Person	0.13	0	0.01	0.14	0.12	0.03	1.2
LRapw-het	Person	0.13	0	0.01	0.14	0.12	0.03	1.2
LRi-crve	Person	0.13	0	0.01	0.14	0.12	0.03	1.2

method	weight	mean.SE.hat	bias	MCSE.cover	SE.s	SE.f	MCSE.SE	ratio
LRi-db	Person	0.13	0	0.01	0.14	0.12	0.03	1.2
LRicw-db	Cluster	0.13	0	0.01	0.14	0.12	0.03	1.2
MLM	Cluster	0.14	0	0.00	0.14	0.12	0.03	1.2

Very good match! And even in this fairly extreme context, the difference in SEs is not large at all.